Diep Dao

Results-oriented Software Engineer with 10+ years of experience in designing and implementing AI-powered solutions. Proficient in leveraging cutting-edge technologies to solve complex business problems and deliver impactful outcomes. Skilled in full-stack development, machine learning, and data analytics. Proven ability to drive projects from concept to production, ensuring high-quality and scalable solutions.

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EXPERIENCES

Lead Developer, Standard Chartered Bank — *Singapore*

Dec 2021 - Present, 4 years+

- Working on a financial market terminal (FMT) platform and an in-house apps framework, a core member contributes to an open framework used by teams across the globe (Bangalore/HK/London). Reducing time-to-market for the FM critical apps used by traders/sales/BAs, tech stacks are Java, Guice, Grpc, Websocket, React, Solace, and Hazelcast.
- Implementing real-time risk management apps for trading desks, by computing PnL / PV / DV01 / Greeks. Tech stacks are Java, in-house model engine, Kafka, Hazelcast, ClickHouse, in-house apps framework.
- Collaborating and advising global teams to develop financial apps from scratch to production, by adopting the in-house built framework.
- Developing a process to integrate Machine Learning models into the FMT platform, promoting advanced analytics & Python development across the global teams.
- Technical mentor for fresh graduates and internships, lead the Generative AI prototype developments.

Senior System Analyst / Machine Learning Engineer, DHL Express — *Singapore*

Apr 2021 - Dec 2021, 9 months

- Built and deployed end-to-end Machine Learning projects, 2 in production and 1 in pilot within 9 months, reduced the budget by 70%. Worked closely with data scientists, solution architects, and engineers from Malaysia, India, and Germany to deploy and integrate ML APIs into the global platform.
- Implemented the ML project development template used for the Business Intelligence Center of Excellence, tech stacks are Python/Java/R, MLFlow, Airflow, Teradata, Openshift, Splunk, and Jenkins.
- Refactored and optimized POC/Pilot code, wrote unit/integration tests, and developed APIs/batch-scoring services for ML Models. Implemented data ingestion pipelines, AutoML, and CI/CD pipelines. Created monitoring dashboards, alerts, and sent scheduled model prediction/classification reports to business users.

Senior NLP Data Scientist, Refinitiv / London Stock Exchange — *Singapore*

Feb 2018 - Mar 2021, 3 years 2 months

- Built and deployed the first-ever Deep Learning (LSTM) based model to production for the Intelligent Tagging platform. Implemented the POC in Python, then ported the model to Java/Scala for low latency with optimized code and unit tests, created CI/CD, and integrated with the main framework.
- Led the regional NLP Data Scientist team by implementing a common framework for multiple NLP projects, reducing the POC timeline. Set up coding guidelines, coding review, and mentoring team members.
- Coordinated with global teams (Beijing, London, Bangalore) to acquire business requirements, implemented POCs with demos, built and deployed end-to-end ML/DL models to production using Python/Scala, Keras/Tensorflow.
- Implemented Chinese language capabilities for the Media Check (KYC, due diligence) product by implementing an entity-relationship model to detect risk events (AML, corruption, terrorists) in news media using Logistic Regression / Bidirectional LSTM models.

Senior Data Engineer, Sentifi — *Ho Chi Minh, Vietnam*

Mar 2015 - Feb 2018, 3 years

- Sentifi is a Swiss-based fintech startup focusing on crowd intelligence from social networks. Where I helped build and deploy the first-ever Machine Learning model for the data team, reduced 90% manual work for a keyword-based system, and increased the accuracy by 60% for the Message Scorer, measuring how relevant a text is to the financial domain, using the Naive Bayes model, TF/IDF features, Scikit-learn and Python.
- Developed scalable ETL components for a near-real-time system streaming billions of messages from Twitter, news, and blogs using frameworks built in-house: Python/Java/Scala, SparkJava, RabbitMQ, Elasticsearch, and Docker.
- Increased the message throughput by 30 times for the Text Similarity Checker component in the streaming system by implementing the MinHash algorithm.

Software Engineer, Rivercrane — *Ho Chi Minh, Vietnam*

Mar 2014 - Feb 2015, 1 year

- Developed features for a Business Intelligence web application that analyzes revenue/profit/budget from millions of e-commerce transactions. Full-stack development from optimizing SQL queries, writing backend services using Java Spring to developing analytic UIs using JavaScript, jqGrid, and HighCharts.

EDUCATION

Computer Science, Bachelor of Engineering — *Ho Chi Minh City University of Technology*

Sep 2010 - Dec 2014

Honor class (10%), GPA 8.2/10.

SKILLS

Languages: Java (fluent), Python (fluent), Scala / Javascript / HTML / CSS (prior experience), Git / Bash, and Linux.

Full Stack Development: Spring, Flask, Guice, Bootstrap, React. Server-side apps development.

Databases: PostgreSQL, Elasticsearch, Cassandra, ClickHouse and Redis.

Machine Learning / Deep Learning, Natural Language Processing with Tensorflow, Keras, and Scikit-learn.

Big Data, Parallel Processing and Distributed Systems with Spark and Hadoop. Data Pipeline / ETL, Batch & Stream Processing.

Docker, Microservices, AWS and Google Cloud Services.

Cryptography, Blockchain and Smart Contract Development for DeFi / Tokenized Assets.

CERTIFICATES

Blockchain Specialization

AWS Certified Solutions Architect

Deep Learning Specialization

Machine Learning

Big Data Analysis with Apache Spark

Introduction to Apache Spark

LANGUAGES

English
(Professional Working Proficiency).

Vietnamese
(Native Proficiency).

INTERESTS

Math & sciences.

Documentaries and histories.

Playing tennis.